

Frank Hu

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Stanford, CA 94305-5080, USA

RESEARCH INTERESTS

Generative AI and foundation models (transformers, large language models, chemical language models, vision language models, diffusion models, flow matching models) for molecule and protein design; gradient-based post-training algorithms for molecular optimization and inverse design problems; physics-informed neural networks for molecular and biomolecular simulations and modeling.

EDUCATION

Stanford University

PhD Candidate in Chemistry, GPA: 4.00/4.00

Stanford, CA

August 2022 - Present

Carnegie Mellon University

MS in Chemistry, GPA: 3.97/4.00

BS in Chemistry, GPA: 3.97/4.00

Pittsburgh, PA

August 2020 - August 2022

August 2018 - August 2022

RESEARCH

Markland Lab and Rotskoff Lab, Department of Chemistry, Stanford University

Graduate researcher with Professor Thomas E. Markland and Professor Grant M. Rotskoff

Stanford, CA

August 2022 - Present

- Building foundational chemical language models and machine learning frameworks for small-molecule design, optimization, and conformational sampling
- Building chemical language model-based frameworks for direct interpretation of spectroscopic data
- Developing data-efficient machine-learned force fields for condensed phase molecular dynamics

Prescient Design/AI4DD, Genentech

Machine learning research intern with the foundation models team

South San Francisco, CA

June 2026 - September 2026

- Large-scale asynchronous reinforcement learning for adapting open-weight LLMs to small-molecule design
- Assessing the sample efficiency of frontier LLMs on continuous and discrete batch optimization settings

Yaron Lab, Department of Chemistry, Carnegie Mellon University

Undergraduate and Master's researcher with Professor David J. Yaron

Pittsburgh, PA

August 2018 - August 2023

- Built a novel framework for machine-learned parameterization of semiempirical chemical Hamiltonians
- Thesis title: *DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians*

Jiang Lab, Department of Mathematics and Statistics, Georgia State University

Research intern with Professor Yi Jiang

Atlanta, GA

June 2017 - April 2018

- Developed computational methods for analyzing cancer cell metastasis based on microscopy image data

PUBLICATIONS AND PRESENTATIONS

Publications

- 1 **F. Hu***, S. Chennakesavalu*, Z. Wang, P. Suriana, B. Vani, K. Shmilovich, K. Chuang, C. Grambow. Training Large Language Models for Small-Molecule Design with Synthetic Task Scaling. *arXiv preprint, arXiv:2609.04735*. (2026)
- 2 **F. Hu**, S. Chennakesavalu, D. Graff. Frontier LLMs are effective batch optimizers: Assessing reasoning models in continuous and discrete settings. *arXiv preprint, arXiv:2609.03177*. (2026)
- 3 S. Ibarra, S. Chennakesavalu, **F. Hu**, G. M. Rotskoff. Efficient, Few-Shot Directed Evolution with Energy Rank Alignment. *J. Chem. Inf. Model.*, **66**, 17, 10609–10621 (2026)
- 4 N. I. Hausman, J. Kelly, M. S. Chen, **F. Hu**, A. Lee, A. Montoya-Castillo, G. S. Schlau-Cohen, T. E. Markland. Streamlining analysis and design of two-dimensional electronic spectroscopy using machine learning. *J. Chem. Phys.*, **165**, 074109 (2026)
- 5 J. Liu, S. Ibarra, **F. Hu***, A. Park*, A. Dunn, G. M. Rotskoff. Generative design of intrinsically disordered protein regions with IDiom. *ICML 2026 Workshop on Generative and Agentic AI for Biology* (2026)
- 6 J. K. Rasch, A. L. Clayborn, D. Sole-Barber, S. Wan, C. A. Saavedra, A. J. Fairhall, **F. Hu**, J. Bartz, Y. Podorova, A. J. Bergsten, Z. D. Young, I. R. Barrera, T. H. K. Mone, N. Asgari, T. E. Markland, R. H. Goldsmith. Label-Free Optical Differentiation of Single Diffusing Amino Acids at Picomolar Concentrations. *Under review, Nature Nanotechnology* (2026)

- 7 S. Ibarraran, S. Chennakesavalu, **F. Hu**, G. M. Rotskoff. Efficient, Few-shot Directed Evolution with Energy Rank Alignment. *ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2026)
- 8 **F. Hu**, J. M. Tubb, D. Argyropoulos, S. Golotvin, M. Elyashberg, G. M. Rotskoff, M. W. Kanan, T. E. Markland. Pushing the Limits of One-Dimensional NMR Spectroscopy for Automated Structure Elucidation Using Artificial Intelligence. *J. Chem. Inf. Model.*, **66**, 12, 6992–7002 (2026)
- 9 S. Chennakesavalu, **F. Hu**, S. Ibarraran, G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. *Conference on Neural Information Processing Systems* (2025)
- 10 J. Kelly, **F. Hu**, A. Damiani, M. S. Chen, A. Snider, M. Son, A. Lee, P. Gupta, A. Montoya-Castillo, T. J. Zuehlsdorff, G. S. Schlau-Cohen, C. M. Isborn, T. E. Markland. Two-Dimensional Electronic Spectroscopy in the Condensed Phase Using Equivariant Transformer Accelerated Molecular Dynamics Simulations. *J. Phys. Chem. Lett.*, **16**, 22, 5561–5569 (2025)
- 11 S. Chennakesavalu, **F. Hu**, S. Ibarraran, G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2025)
- 12 **F. Hu**, M. S. Chen, G. M. Rotskoff, M. W. Kanan, T. E. Markland. Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning. *ACS Cent. Sci.*, **10**, 11, 2162–2170 (2024)
- 13 S. Chennakesavalu, S. K. Manikandan, **F. Hu**, G. M. Rotskoff. Adaptive nonequilibrium design of actin-based metamaterials: Fundamental and practical limits of control. *Proc. Natl. Acad. Sci.*, **121**, 8, e2310238121 (2024)
- 14 P. Eastman, R. Galvelis, R. P. Peláez, C. R. A. Abreu, S. E. Farr, E. Gallicchio, A. Gorenko, M. M. Henry, **F. Hu**, J. Huang, A. Krämer, J. Michel, J. A. Mitchell, V. S. Pande, J. PGLM Rodrigues, J. Rodriguez-Guerra, A. C. Simmonett, S. Singh, J. Swails, P. Turner, Y. Wang, I. Zhang, J. D. Chodera, G. De Fabritiis, T. E. Markland. OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. *J. Phys. Chem. B*, **128**, 1, 109–116 (2024)
- 15 **F. Hu**, F. He, D. J. Yaron. Treating Semiempirical Hamiltonians as Flexible Machine Learning Models Yields Accurate and Interpretable Results. *J. Chem. Theory Comput.*, **19**, 18, 6185–6196 (2023)

Presentations

- 1 (Contributed talk) Climbing the Combinatorial Mountain: Using Artificial Intelligence to Push the Limits of 1D NMR for Structure Elucidation. SMASH 2026, Breckenridge, CO
- 2 (Poster) Generative design of intrinsically disordered protein regions with IDiom. ICML 2026 Workshop on Generative and Agentic AI for Biology, Seoul, South Korea
- 3 (Poster) Efficient, Few-shot Directed Evolution with Energy Rank Alignment. ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design, Rio de Janeiro, Brazil
- 4 (Invited talk) Using AI to push the limits of 1D NMR for structure elucidation. Novartis Graduate Colloquium 2026, Novartis, Cambridge, MA
- 5 (Poster) Pushing the Limits of One-Dimensional NMR Spectroscopy for Automated Structure Elucidation Using Artificial Intelligence. Berkeley Statistical Mechanics Meeting 2026, University of California, Berkeley, CA
- 6 (Poster) Aligning Transformers with Continuous Feedback via Energy Rank Alignment. Conference on Neural Information Processing Systems (NeurIPS) 2025, San Diego, CA
- 7 (Poster) Accurate and Efficient Structure Elucidation from NMR Using Artificial Intelligence. SMASH 2025, Porto, Portugal
- 8 (Contributed talk) Automated structure elucidation from NMR using artificial intelligence. Young WATOC (World Association of Theoretical and Computational Chemists) 2025, Oslo, Norway
- 9 (Poster) Energy rank alignment. ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design, Singapore
- 10 (Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning. Experimental Nuclear Magnetic Resonance Conference 2025, Pacific Grove, CA
- 11 (Poster) Structure elucidation from NMR using machine learning. Berkeley Statistical Mechanics Meeting 2025, University of California, Berkeley, CA
- 12 (Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning. American Conference on Theoretical Chemistry 2024, University of North Carolina, Chapel Hill, NC

- 13 (Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning. West Coast Theory Meeting 2024, University of California, Merced, CA
- 14 (Contributed talk) Semiempirical Hamiltonians as Flexible Model Forms for Machine Learning. ACS 2023, San Francisco, CA
- 15 (Poster) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties. West Coast Theory Meeting 2023, University of California, Davis, CA
- 16 (Contributed talk) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties. Beckman Symposium 2022, Rutgers University, NJ

HONORS AND AWARDS

John Stauffer Memorial Fellowship <i>Stanford University, Stanford, CA</i>	2026
Center for Molecular Analysis and Design (CMAD) Fellowship <i>Stanford University, Stanford, CA</i>	2024-2026
Poster Prize <i>SMASH 2025, Porto, Portugal</i>	2025
Poster Prize <i>West Coast Theory Meeting 2024, University of California, Merced, CA</i>	2024
Poster Prize <i>West Coast Theory Meeting 2023, University of California, Davis, CA</i>	2023
NSF GRFP Honorable Mention <i>National Science Foundation</i>	2023
College Chemistry Award <i>Society of Analytical Chemists of Pittsburgh</i>	2022
Beckman Scholar <i>Arnold and Mabel Beckman Foundation</i>	2021-2022
National Merit Scholar <i>National Merit Scholarship Corporation</i>	2018-2022

LEADERSHIP AND SERVICE

Stanford Graduate Student Affairs Committee Industry Partnerships Co-Chair <i>Stanford University, Stanford, CA</i>	July 2025 - March 2026
◦ Networking and outreach with industry representatives to boost campus recruiting for chemistry PhD students	
Stanford FAST Mentor <i>Stanford University, Stanford, CA</i>	October 2024 - March 2026
◦ Scientific outreach work and mentoring underprivileged high school students doing science fair projects	
Teaching Assistant <i>Department of Chemistry, Stanford University, Stanford, CA</i>	August 2022 - May 2023, January 2025 - March 2025
◦ TA for 4 quarters: general chemistry, statistical mechanics, quantum chemistry, and computational chemistry	
Head Teaching Assistant <i>School of Computer Science, Carnegie Mellon University, Pittsburgh, PA</i>	August 2021 - August 2022
◦ Head TA for an introductory computer science course	
◦ Developed course assignments and exams and arranged TA staff logistics in addition to teaching	